Region4Web: Rethinking Observation Space Granularity for Web Agents

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Paper version: arXiv 2605.07134v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Existing work treats observation at the same element-level granularity as the action space, leaving the page's functional organization implicit.*

Filtering/truncation stay **element-level**; functional layout is never made explicit.

Why regions: two Mind2Web measurements

1. **Actions cluster.** LCA depth 0.48 vs 0.22.
2. **Pages sit still.** 52.9% of steps: zero change.
3. **Yet agents rebuild.** Full AXTree each step.

Evaluation setup

WebArena · 812 tasks | BrowserGym · live OSM | 30-step budget | SteP / AgentOccam | GPT-4o evaluator | no seeds reported

- **All online** WebArena success; no offline metrics.
- **No seeds or variance**; evaluator swapped to GPT-4o.

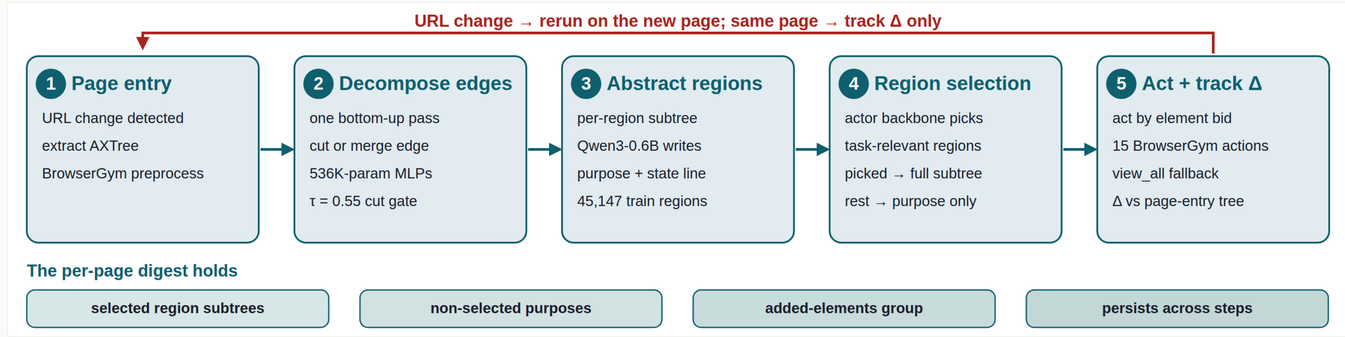
Our read (CUA Radar)

64	★★	transfer
score of 100	band	evidence level

- **Reports:** online WebArena (812); held-out pages; SteP + AgentOccam; 30-step budget; token cost; ablation.
- **Not reported:** variance; evaluator check; code. **Flags:** AP18 lossy context management.
- Pattern C13 (prune/restructure text obs). Authors' numbers, not rerun.

Method: decompose, abstract, select, track

- **Sees** the page AXTree; **acts** by bid, 15 BrowserGym actions.
- **Region4Web:** 536K-param cut/merge edge pass; Qwen3-0.6B writes purpose + state summary per region.
- **PageDigest:** backbone picks regions at page entry; within-page, only the Δ is tracked.



Our redraw (not the paper's figure): page-entry pipeline and the within-page digest loop.

$$\hat{y}_{v,c} = \mathrm{EdgeClassifier}(x_v,\ r_c,\ \bar{r})$$

Cut score per edge: node features, child region vector, sibling mean

Key results: WebArena online success (%)

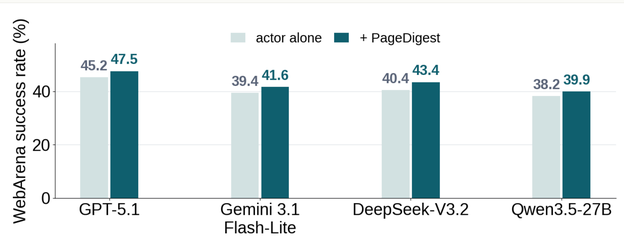


Table 1, redrawn: overall success per backbone, with and without PageDigest; no variance reported.

Obs. tokens −43% avg (6,437 → 3,671); success +2.3%p avg

SO WHAT → *Token counts per step under the o200k_base tokenizer.*

Ablation: WebArena-Lite success (%)

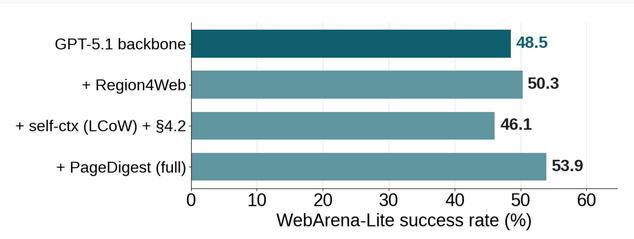


Table 2, redrawn: GPT-5.1 on the 165-task subset.

SO WHAT → *Region4Web alone lifts SR to 50.3; the element-level variant falls to 46.1; PageDigest tops at 53.9 (+5.4%p) on 30% fewer obs. tokens. Obs. length runs 5,410 / 5,922 / 4,013 / 3,814; the element-level variant cuts 26% but drops SR. Failure audit: 82.0% backbone capacity vs 8.0% PageDigest (multi-cause).*

Headline Numbers

43%
avg obs. token cut
6,437 → 3,671 avg/step

+2.3%p avg success gain

47.5% GPT-5.1 + PageDigest

45,147 region annotations

The paper's controls

CONTROL	RESULT	READING
Component ablation	+Region4Web 50.3, full 53.9	element variant 46.1
Agent methods, GPT-4o	SteP 38.7 · Occam 41.3	obs −50% / −16%
Held-out pages	train: Tranco top-1M	WebArena self-hosted
Variance · evaluator	no seeds or std	GPT-4o swap untested

Takeaway

Functional regions + a per-page digest: fewer tokens, equal-or-better success, policy untouched.

2,052 verified pages | Qwen3-0.6B abstractor | median task tokens −25%

Scan to read the full paper

arxiv.org/abs/2605.07134v1